Daily Geometry Drill #7

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. Write down the locus of points whose distance from the origin is 5.

Ans: $x^2 + y^2 = 25$

2. Write down the locus of points equidistant from the two points $(0, 0)$ and $(4, 0)$.

Ans: $x = 2$

3. The locus of P with $PA = 2\,PB$, where $A = (0, 0)$ and $B = (6, 0)$, is a circle. It crosses the segment AB at one point. Write down the x-coordinate of that point.

Ans: 4

4. Write down the locus of points that are always 3 units from the fixed line $y = 4$.

Ans: $y = 1$ or $y = 7$

5. The locus of points whose distance to the point $(0, 1)$ equals their distance to the line $y = -1$ is a parabola. Write down its equation.

Ans: $x^2 = 4y$

6. The locus of points whose distance to the point $(1, 2)$ is exactly 3 is a circle. Write down its equation.

Ans: $(x-1)^2 + (y-2)^2 = 9$

7. Write down the locus of the midpoint M of every radius OQ of the circle $x^2 + y^2 = 4$ as Q runs around the circle (with O the origin).

Ans: $x^2 + y^2 = 1$

8. The equation $x^2 + y^2 = 2x$ describes a circle. Write down its radius.

Ans: 1

9. Write down whether the statement "$x^2 + y^2 = 4$ implies $x = \pm 2$ or $y = \pm 2$" is true or false.

Ans: false

10. Write down the locus of points P with $PA = PB$, where $A = (1, 3)$ and $B = (5, 3)$. (Give its equation in x.)

Ans: $x = 3$

Section B Manipulation Drills

1. The locus of P with $PA = 3\,PB$, where $A = (0, 0)$ and $B = (8, 0)$, is a circle. What is its radius? *(after tmua.fyi Challenge Mock 1 Paper 1 Q14)*

A) 9

B) 6

C) 3

D) 8

Ans: C) 3

2. Locus of points equidistant from $(0, 0)$ and $(0, 4)$ is?

A) $x = 2$

B) $y = 2$

C) $y = 4$

D) $x = 4$

Ans: B) $y = 2$

3. The points satisfying $x^2 + y^2 \leq 4$ form a filled disk. Its area is?

A) 2π

B) 4π

C) 8π

D) 16

Ans: B) 4π

4. The points $A = (0, 0)$ and $B = (4, 0)$ are fixed. The point P moves in the region $y > 0$ so that $\angle APB = 45°$. What is the locus of P?

A) an arc of the circle with centre $(2, 0)$ and radius 2

B) an arc of the circle with centre $(2, -2)$ and radius $2\sqrt{2}$

C) part of the line $x = 2$

D) an arc of the circle with centre $(2, 2)$ and radius $2\sqrt{2}$

Ans: D) an arc of the circle with centre $(2, 2)$ and radius $2\sqrt{2}$

5. A point is known to lie strictly inside the unit disk $x^2 + y^2 < 1$. Which of the following statements is *necessarily* true for such a point?

A) $x^2 + y^2 < \dfrac{1}{2}$

B) $x + y < 1$

C) $x^2 + y^2 < 2$

D) $x^2 + y^2 > 0$

Ans: C) $x^2 + y^2 < 2$

6. The Apollonius locus $PA = 3PB$ with $A = (0, 0)$, $B = (2, 0)$: what is the x-coordinate of its right endpoint on the x-axis?

A) 3

B) 4

C) 6

D) 9

Ans: A) 3

7. To prove a point P lies on the unit circle $x^2 + y^2 = 1$, which single condition is *sufficient* on its own?

A) $x^2 + y^2 < 1$

B) $x = -\dfrac{1}{2}$ and $y = \dfrac{\sqrt{3}}{2}$

C) P is not the origin

D) $x + y = 1$

Ans: B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$

8. Which property is necessary, but not sufficient, for a quadrilateral to be a rectangle?

A) all four angles are right angles

B) the diagonals are equal in length

C) the diagonals are perpendicular

D) the diagonals are equal in length and bisect each other

Ans: B) the diagonals are equal in length

9. A circle has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Write down its centre.

Ans: $(2, 3)$

10. Write down the radius of the circle $x^2 + y^2 - 4x + 6y + 9 = 0$.

Ans: 2

Section C Substitution & Structure

1. The locus of P with $PA = 2\,PB$, where $A = (-2, 0)$ and $B = (4, 0)$, is a circle. What is its centre?

A) $(6, 0)$

B) $(8, 0)$

C) $(10, 0)$

D) $(2, 0)$

Ans: A) $(6, 0)$

2. The locus of points equidistant from the lines $L_1 : x + y = 0$ and $L_2 : x - y = 0$ is:

A) $y = x$

B) $y = -x$

C) $y = |x|$

D) $x = 0$ or $y = 0$

Ans: D) $x = 0$ or $y = 0$

3. *Claim.* If A, B and P lie on a circle with centre O, and P is on the major arc AB, then $\angle AOB = 2\angle APB$.

Proof. Draw the diameter PD. Triangles OPA and OPB are isosceles, so by the exterior angle theorem $\angle AOD = 2\angle APO$ and $\angle BOD = 2\angle BPO$. Adding, $\angle AOB = 2(\angle APO + \angle BPO) = 2\angle APB$.

Which statement about this argument is correct?

A) It is correct and covers every position of P on the major arc.

B) The exterior angle step fails, because $\angle OAP \neq \angle OPA$ in general.

C) Adding assumes O lies inside $\angle APB$; when it does not, the angles must be subtracted, so that case is unproved.

D) The claim itself is false when O lies outside triangle APB.

Ans: C)

4. $PQRS$ is a quadrilateral. Consider the conditions: I. its diagonals bisect each other; II. its diagonals are equal in length; III. its diagonals are perpendicular. Which is the smallest set of these conditions that together guarantee $PQRS$ is a square? *(after TMUA 2020 Paper 2 Q7)*

A) I and II

B) I and III

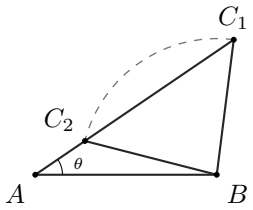
C) II and III

D) I, II and III

E) no set of them is enough

Ans: D) I, II and III

5. In $\triangle ABC$, $AB = 8$, $BC = 6$ and $\angle A = \theta$. Exactly two non-congruent triangles fit these data, and the larger has four times the area of the smaller. What is $\cos\theta$? *(after TMUA 2018 Paper 1 Q19)*



A) $\dfrac{5\sqrt{7}}{16}$

B) $\dfrac{3\sqrt{14}}{16}$

C) $\dfrac{\sqrt{7}}{4}$

D) $\dfrac{3}{4}$

Ans: A) $\frac{5\sqrt{7}}{16}$

6. The circles $C_1 : (x+3)^2 + (y-1)^2 = 25$ and $C_2 : (x-8)^2 + (y-1)^2 = 9$ have radii 5 and 3. How many common tangents do they have?

A) 0

B) 1

C) 2

D) 3

E) 4

Ans: E) 4

7. The point P moves so that $PA = 3\,PB$, where $A = (0,0)$ and $B = (6,0)$. What is the greatest possible value of PA?

A) 9

B) 7.5

C) 12

D) 6

Ans: A) 9

8. Fil finds the locus of P with $PA = 3\,PB$, where $A = (0,0)$ and $B = (2,0)$, by writing $(x-2)^2 + y^2 = 9$. What is wrong?

A) Squaring loses sign information, so the locus is really a single point.

B) The locus of $PA = 3\,PB$ is a straight line, not a circle.

C) His equation says $PB = 3$; it never uses the distance to A.

D) Nothing: $(x-2)^2 + y^2 = 9$ is the locus.

Ans: C)

Section D Challenge Ramp

1. The locus of P with $PA^2 + PB^2 = 34$, where $A = (-3, 0)$ and $B = (3, 0)$, is a circle. What is its radius?

A) $2\sqrt{2}$

B) 4

C) $\sqrt{17}$

D) $\sqrt{34}$

Ans: A) $2\sqrt{2}$

2. S is the square with vertices $(0,0)$, $(4,0)$, $(4,4)$ and $(0,4)$. What is the area of the region of points whose distance from the boundary of S is at most 1?

A) $32+\pi$

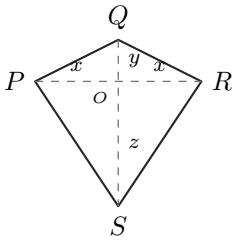
B) $28+4\pi$

C) $23+\pi$

D) $28+\pi$

Ans: D) $28+\pi$

3. The diagram shows a kite $PQRS$ whose diagonals meet at O, with $OP = OR = x$, $OQ = y$ and $OS = z$. Which condition is necessary and sufficient for P, Q, R and S to lie on a circle? *(after TMUA 2022 Paper 2 Q11)*



A) $x = y = z$

B) $2x = y + z$

C) $x^2 = yz$

D) $y = z$

E) $y^2 = x^2 + z^2$

Ans: C) $x^2 = yz$

4. The circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x-4)^2 + y^2 = 9$ meet at P and Q. What is the length PQ?

A) 3

B) 6

C) $\dfrac{4\sqrt{14}}{3}$

D) $\dfrac{24}{5}$

Ans: B) 6

5. The points A and B are fixed, and $k > 0$ with $k \neq 1$. Prove that the locus of points P with $PA = k\,PB$ is a circle, and that this circle cuts every circle through A and B at right angles.

Ans: Proof: see method